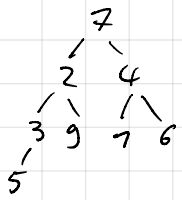
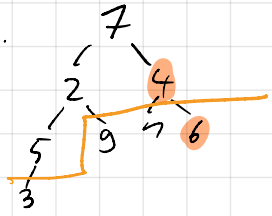
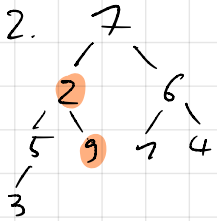
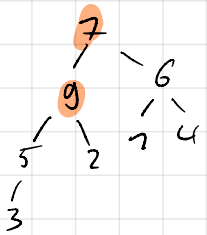
A 4.2 7 2 4 3 9 1 6 5

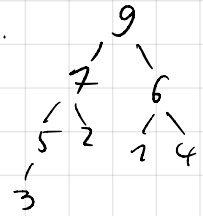


Erstelle MaxHeap

1. 

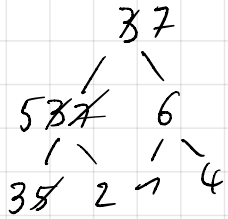
2. 

3. 

4. 

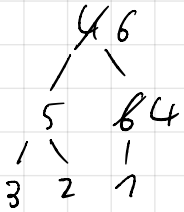
9 7 6 5 2 1 4 3

1. Heap abbauen 3 7 6 5 2 1 4 | 9



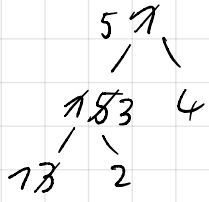
7 5 6 3 2 1 4 | 9

2. Heap abbauen 4 5 6 3 2 1 | 7 9



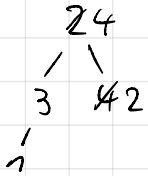
6 5 4 3 2 1 | 7 9

3. Heap abbauen 1 5 4 3 2 | 6 7 9



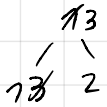
5 3 4 1 2 | 6 7 9

4. Heap abbauen 2 3 4 1 | 5 6 7 9



4 3 2 1 | 5 6 7 9

5. Heap abbauen 1 3 2 | 4 5 6 7 9



3 1 2 | 4 5 6 7 9

6. Heap abbauen 2 1 | 3 4 5 6 7 9



2 1 | 3 4 5 6 7 9

7. Heap abbauen 1 | 2 3 4 5 6 7 9



1 2 3 4 5 6 7 9

A 4.3

1)

```
n =       100
measurement  1 (generated):     0,027 ms.
measurement  2 (generated):     0,029 ms.
measurement  3 (generated):     0,029 ms.
measurement  4 (generated):     0,029 ms.
measurement  5 (generated):     0,028 ms.
average:    0,028 ms.

n =      1000
measurement  1 (generated):     0,270 ms.
measurement  2 (generated):     0,416 ms.
measurement  3 (generated):     0,285 ms.
measurement  4 (generated):     0,269 ms.
measurement  5 (generated):     0,327 ms.
average:    0,313 ms.

n =     10000
measurement  1 (generated):     3,655 ms.
measurement  2 (generated):     2,822 ms.
measurement  3 (generated):     3,853 ms.
measurement  4 (generated):     2,657 ms.
measurement  5 (generated):     2,714 ms.
average:    3,140 ms.

n =    100000
measurement  1 (generated):    28,851 ms.
measurement  2 (generated):    31,846 ms.
measurement  3 (generated):    42,891 ms.
measurement  4 (generated):    31,904 ms.
measurement  5 (generated):    31,337 ms.
average:   33,366 ms.

n =   1000000
measurement  1 (generated):   342,251 ms.
measurement  2 (generated):   220,697 ms.
measurement  3 (generated):   218,364 ms.
measurement  4 (generated):   216,588 ms.
measurement  5 (generated):   214,748 ms.
average:  242,530 ms.

n =  10000000
measurement  1 (generated):  2744,229 ms.
measurement  2 (generated):  2731,883 ms.
measurement  3 (generated):  2860,267 ms.
measurement  4 (generated):  2439,759 ms.
measurement  5 (generated):  2311,900 ms.
average: 2617,608 ms.

- done -
```

2) Lineares Wachstum $O(n)$

A 4.5

